Stage 5. Bills

Bill validation rules, imported and manual bills, images, notes and approval

PREPARED BY Energy Hippo implementation team
STATUS Ready to enter (revised 2026-11-10)

Stage 5 puts two years of real utility bills into Northlake: one monthly bill per account from January 2024 through December 2025, decomposed into the charge lines each provider actually bills, plus the fixtures that make Bill Processing and Bill Tracking reports worth looking at - a meter swap mid-bill, an estimate that gets corrected, a file imported twice on purpose, an account that goes quiet, a rate that changes at a year boundary, and a file that a bad fiscal date bounces. Bill Validation rules go in first so the deliberate failures actually fire when the bills that should trip them are saved. Stage 2's hierarchy and Stage 4's rates are prerequisites; nothing here needs Stage 3's interval data, though the interval-metered buildings' demand lines read better once it exists.

Who enters what (reconciled 2026-11-10). Bill Entry lists only Bill Entry accounts, so the recipe now says one thing everywhere. The 31 Valley Electric and River City accounts, Cedar Row Apartments' and the Thermal Plant's included, are Bill Importer accounts and nora.chen imports them. The 13 Sierra Gas and Helios accounts are Bill Entry accounts: iris.morales types them, and marcus.reed types Cedar Row's own gas and PPA accounts. The six Thermal Plant cost centers are Bill Allocation accounts, so Bill Entry cannot open them and their bills arrive with Stage 7's allocation (keiko.tan reviews); Ridgeline's own statement is generated, also in Stage 7. That makes 44 accounts and 1,049 bills in this stage (1,193 once Stage 7 adds the 144 plant bills, 1,211 with Ridgeline's 18 statements).

Added 2026-10-01: Ridgeline Wind. The Thermal Plant's wind company brings two accounts, and this stage enters only one of them. Valley Electric's generator account at the Ridgeline substation (SYN-VED-A-0010014) is imported like every other Valley Electric account, but it opened with the acquisition. Valley Electric reads on the 5th, so its first bill runs 35 days, from the July 1 opening to August 4, and its series is 17 bills (August 2024 to December 2025), not 24. It keeps none of its export credit: the credit leaves each generator bill as a transfer line and lands, sign reversed, on the Central Plant's bill one cycle later (the first bill's credit on the September 2024 bill), so the generator pays a small residue and the credit is counted once. Ridgeline's own transfer agreement (SYN-RWL-G-0000001) takes no bill in this stage: EEM generates its monthly statement in Stage 7, and the hand-made July 2024 statement is the figure the first generated one must match to the cent. Three itemized Valley Electric statements and that hand-made one are in `documents/utility-bills`; each step below says what they change.

Before you start

- Stage 1's companies, providers, calendars, groups and users are in place; Stage 2's billing accounts and meters exist with their synthetic account and meter numbers recorded, the Ridgeline generator account and its interconnection meter included; Stage 4's reworked rates are loaded - VED-TOU-GS with its 2025-01-01 price date (there is no separate FY25 rate), VED-TOU-PRI, and the generator rates VED-GEN-IC and VED-BCT.

- Accounts and meters are referenced here by synthetic number, not by database ID - look each one up in Stage 2's records before opening it.
- Before Step 5, confirm the approver holds the Bill Validation function: that is what the product checks when a bill is validated, and Stage 1 grants it through EEM Company Setup (see Findings).
- Leave SysConstants IsValidateBillsJob unset until the end of Step 5. The validation job approves and un-approves bills by itself once it is on (see Step 5 and Findings).
- Step 3's bills carry no per-line Expense Account yet: that is Stage 6 (`per_line_coding`), after the GL accounts exist.

Step 1. Bill Validation rules

Where: Database Administrator, Nodes, the company node's Bill Validation tab (the same page opens from the main window's Bill Validation item).

Enter: 01-bill-validation-rules.yaml, twelve records: the Global defaults, the Northlake University Company Defaults, the Validation Bay group (created in Manage Groups, then its own settings), seven account records (the five planned overrides, the Ridgeline generator's holding two rows, plus two "watch" accounts, Admin Hall and the Library), and `test_catalog`, which lists all 48 tests.

The test list. The product has 47 numbered tests and a 48th, number 99 (accrual bill without a mapping). Their numbers and meanings are the index at the top of `BillValidation_Run.sql`; each is one `BillValidation_Calc*` procedure, and `test_catalog` says for every test which calculation it reads, where it is turned on, a bill that trips it and a bill that does not. 39 are exercised that way. Nine are not, each with its reason (6 no account is closed, 9 the editor repairs a reversed period, 29 needs a year of demand history, 37 and 38 need a Baseline bill scenario, 42 no bill bills the one closed meter, 44 needs a fall against a known year, 45 and 46 need a Weather Normalized scenario); 6, 9 and 42 are turned on anyway so they report the day a bill can trip them.

Levels. The screen has four scopes - Global, Company, Group and Billing Account - and no provider scope. Global holds the planned families (cost change, cost maximum, usage change, usage floor, period length, meter open and closed). Northlake University's Company Defaults hold the date and presence tests that never report on a clean bill (previous balance, estimated bill, reconciliation, billing before account opening, the date orders, duplicate period, invoice delay, unmapped accrual). The Validation Bay group (Lakeview Hall's gas account) holds the tests whose deliberate failures are fixture bills that are read and deleted. Two accounts, Admin Hall and the Library, carry the electric usage, demand and price tests that the sample PDFs trip.

Both tabs. The earlier text said Northlake uses Bill Save only. Bill Save runs when a bill is saved in Bill Entry and stores nothing; an imported bill is never saved in the editor, so it is validated only by the validation job, which runs the Batch Test tab and writes BillValidation_Tbl. So the Global, Company and account rows are entered on both tabs, and the Stage 6 approval queue reads what the job stored.

Order inside the editor: pick "Validation for" (Global Test Defaults, or a company), then "Settings for" if you picked a company (Company Defaults, Manage Groups, a Group or one billing account). Test Name, Commodity and the paired description are locked from the product's own catalog; State, Percent, Amount and Unit are yours to set, and Percent and Amount only where the test has that cell. A Company, Group or Account row replaces the whole configuration of the row it overrides, so the Science Center override repeats the 1,000 dollar floor and Global test 25 now carries a minimum usage (without one it could never report).

The overrides each open one billing account through "Settings for > Account": Science Center's cost-change threshold is 60% (lab loads swing further than the 35% campus default), Student Center's gas usage-change

test is turned off (the estimate-then-correction fixture is a deliberate swing), Cedar Row A's minimum-period-length test is turned off (the July 2024 renewal produces one short period), the Thermal Plant's Science Center allocation gets its own $250,000 cost ceiling (entered under Northlake University: all six cost centers belong to it, not to the Thermal Plant company as the earlier text said), and, added for Ridgeline, the Valley Electric generator account has both of its tests against the previous bill (cost change and electric usage change) turned off. Its bill is the small residue left after the export credit leaves ($1,077.45 on the first bill, $6,753.10 on the calm bill for January 5 to February 4, 2025) and its only usage, station service, follows the wind (4,428 kWh, then 46,920 kWh). The percent column holds at most 255, so raising the threshold could not cover a swing that size.

Record: nothing generated; the scope and test you picked are the record.

Check: `validation_rules_set` (four result sets), then `validation_findings_expected` once the job has run (Step 5).

Step 2. Bill Importer sources

Where: the Bill Importer app, then Web SPA, `#/BillImporter`.

Enter: 02-bill-importer-sources.yaml, two provider sources and two hand-built sample files.

Order: in the Bill Importer app, point it at the provider's sample CSV and let it write a `.bif` file; there is no screen to add a new provider to the app - a provider is a `Schemas/<code>.xml` file plus a code in `appsettings.json`'s utility list, both already installed alongside the app for VED and RCU. Then, in the Web SPA, open Files and upload the `.bif`, open Mappings and set the Account Mapping and Commodity Mapping once per provider, then Workspace to review and import.

The two fixture files skip the WPF step entirely: they are flat BIF text typed by hand in the same column layout the app itself produces, uploaded straight to the Files tab. One declares a Sierra Gas Student Center line in `BTU/hr`, a unit its commodity does not accept. The other is Admin Hall's January 2025 Valley Electric bill keyed with its year transposed, 2052 for 2025, a fiscal period no Northlake calendar has (they run to 2026 or 2027, and EEM keeps extending them) - the Workspace shows the product's own messages, "Fiscal Year NOT found!" and "Fiscal Period NOT found!", and imports nothing.

Added for Ridgeline: seven charge codes, not a source. Valley Electric's generator account and the credit lines on the Central Plant's bills arrive in the same Valley Electric file, so Step 2 only gains mappings: a fixed charge, station service energy, reactive energy, the generation credit, the credit transferred out, the credit received and the BCT fee. VED.xml gets a BillItem block for each. The statement prints the export register in MWh (2,496.612) beside kWh (2,496,612), and the interval channels are kWh. A bill item has no unit of its own, so a quantity takes the unit of the commodity it maps to, and the export is carried as dollars only. Ridgeline Wind itself has no source: nobody sends it a bill.

Record: nothing generated by this step; Step 3's bills carry the IDs.

Check: `imports_configured`, `fiscal_period_rejection_logged`.

Step 3. Bills

Where: Web SPA, `#/BillEntry`, Add Bill, `#/BillEntry/Editor/Details` for every manually-entered account; the imported series from Step 2 for Valley Electric and River City.

Enter: 03-bills.yaml, a monthly series rule per provider group covering the 44 billing accounts that take bills in this stage (the Thermal Plant's six are a reference for Stage 7), six named fixtures, the Ridgeline records described below, the sample totals and header and line fields, and the Validation Bay fixture bills.

Every bill follows its provider's read cycle and posts to the month it ends in: Valley Electric reads on the 5th (a bill runs from the 5th to the 4th), Sierra Gas and River City's Cedar Row cycle on the 15th, and River City's campus cycle, the Thermal Plant and Helios close with the calendar month. The sample bills under `documents/utility-bills/` are exact and itemized, generated from `data/bills/` on those cycles and at the tariffs; the fixtures quote them.

Order inside the editor: period start/end/days, fiscal year/period and Bill Type, then received/issue/due dates and invoice number, then previous balance, payment received, pay amount (Bill Payment accounts only) and total due, then the Bill Items grid - one line per charge from the bill-charge-line spec, in the order the provider bills them, taxes last. A line whose quantity is zero for that period (a winter peak line, an empty water tier, Cedar Row's stormwater line) is left out entirely rather than entered as zero. A bill becomes read-only once it is exported, so enter and check it before that point.

The six fixtures:

- **Meter replacement** - the Library's meter was exchanged on August 15, 2024, inside its September 2024 cycle (August 5 to September 4). It is one bill ($5,114.37) whose items carry each meter's own read period, August 5 to 14 on the old meter and August 15 to September 4 on the new one, so the cycle is billed once, not twice.
- **Estimated then corrected** - Sierra Gas Student Center's November 2024 bill (October 15 to November 14) is entered once with Bill Type Estimated ($935.90), and a second time, independently, with the actual read after the December read ($885.32). There is no Estimate checkbox on a line and no "supersedes" link in the editor, so both bills stay on the account and validation tests 3 and 15 report on them.
- **Duplicate import** - River City's Main Campus file delivered on September 4, 2024 (the campus accounts' August bills) is imported twice on purpose (Step 2), as River City sent it; each account's water, sewer and stormwater lines must still show once.
- **Account number change** - Cedar Row A's River City account was renumbered at the July 2024 lease renewal; its September 2024 bill (August 15 to September 14, $1,984.50) is under the new number, and Stage 2's Previous Account Number links the old one.
- **Late bill** - Sierra Gas Library stops after its November 2025 bill; with no December bill and its usual entry lag past, it is overdue.
- **Rate change boundary** - Valley Electric's FY2025 rates start with service on January 1, 2025, which falls inside Student Center's January 2025 bill (December 5 to January 4). That bill is priced in two parts: energy by the date it was used, demand and the facility charge prorated by days ($8,469.10). The bills either side are wholly at one rate. The 2025 step is a second price date on VED-TOU-GS. The Valley Electric fixtures need item periods, which the flat BIF drops, so they go in as the source service's native XML.

Ridgeline (added 2026-10-01):

- **Generator series** - SYN-VED-A-0010014 gets 17 bills, posting August 2024 through December 2025 and imported with the other Valley Electric accounts. A bill runs from the 5th to the 4th and posts to the month it ends in, so the first runs July 1 to August 4 (35 days) and the last November 5 to December 4,

2025. Each has a customer charge, station service energy, reactive power beyond the 0.95 power-factor allowance (left out when zero), the generation credit, the same credit transferred out, and 5% tax on the three charges. The credit and its transfer cancel, so the bill is a small positive residue. No bill exists before July 1, 2024, and the account has no FY2025 rate step.

- **Credit transfer** - from the September 2024 bill, the Central Plant's bills carry the VED-BCT lines: the credit received, equal to the transfer out on the previous generator bill, and a $12.50 fee that starts with the first credit (an assumption; the rider is effective July 1, 2024, but the September bill is the first one printed). That is 16 credits. The credit is counted once: it leaves one bill and lands on the other. The last generator bill's credit (the one for November 5 to December 4, 2025) has no bill to land on inside the window (it would post to January 2026, after the last bill this packet covers), so it stays in transit. The credit and the fee are not taxable, which is the one place a tax line is not a percent of every line before it.
- **Itemized bills** - like every sample bill, three Ridgeline statements are exact, down to the cent: the generator's first bill (July 1 to August 4, 2024, $1,077.45), its calm bill for January 5 to February 4, 2025 ($6,753.10; station service more than ten times the first bill's, no reactive charge) and the Central Plant's first credit (August 5 to September 4, 2024, $151,171.47). Each is entered three days after its period ends, after its invoice date. Enter them first; their PDFs attach in Step 4.
- **Ridgeline's own statement** is not entered. EEM generates it in Stage 7; the hand-made July 2024 statement (2,251,384 kWh at $0.058 less the $1,250.00 research credit, $129,330.27) is kept in `03-bills.yaml` as the acceptance reference.

Header and line fields. `header_field_bills` and `line_field_bills` set the fields no bill set before, on the bills the sources support: invoice numbers (the pattern the source service prints) and received dates on every Bill Entry bill; a previous balance of 1,310.55, a payment of 500.00 and a pay amount on the Helios Student Center February 2025 bill (Total Due 3,439.55); Bill Type Estimated on the Sierra Gas estimate and the Helios March 2025 backfill bill; Exclude on every Helios rec_transfer line and on each Sierra Gas transport therms line. Provider Meter Number is not a bill field: the grid shows the meter's, set in Stage 2. The per-line Expense Account is Stage 6.

Known totals. `sample_bill_totals` lists the 24 bills whose total is printed on a sample PDF. For every other bill the expected total exists only as the output of the source service's bill generator; `expected_totals_feed` gives the command that writes all 1,193 to `expected-bill-totals.sql`, which `checks.sql` loads (in the same session) and compares bill by bill.

The Validation Bay. `validation_fixture_bay` has twenty small gas bills (F01 to F20) for Lakeview Hall's account that trip the validation tests no real bill trips, each with the tests the editor must list. Enter them after the series, read the findings, delete all twenty.

Record: nothing new beyond the bills themselves; Stage 6 refers back to these accounts by synthetic number.

Check: `bills_per_account`, `charge_lines_sum_to_total`, `bill_totals_expected_vs_actual`, `bill_type_set`, `header_fields_set`, `line_fields_set`, `entered_by`, `estimated_then_corrected`, `meter_replacement_continuity`, `late_bill_present`, `duplicate_file_import`, `account_number_change`, `rate_step_bill`, `generator_account_opened_mid_window`, `bct_credit_counted_once`, `ridgeline_transfer_not_entered_here`, `thermal_plant_not_entered_here`, `validation_fixture_bay_clean`.

Step 4. Bill images and notes

Where: the Bill Entry editor's image pane for bill images; the web Note History screen for a meter note; the Utility Accountant tab's Notes page for a billing-account note.

Enter: 04-bill-images-and-notes.yaml, nineteen of the twenty-four bill images and five notes.

Order: attach each sample PDF to the bill it illustrates - the provider's standard sample to a typical month, the scenario samples to the matching fixture bill. Meter Definition has no Notes page of its own, so the Library meter's replacement note and the Science Center meter's power-factor note both go through Note History with Table Code METR and that meter's ID. Ridgeline adds three Valley Electric statements (the generator's first bill and its calm bill ending February 4, 2025, the Central Plant's first credit), a note on the generator account (transferred at the acquisition, the seller's number SYN-VED-A-0019014) and the hand-made July 2024 Ridgeline statement. Nobody enters that bill, so its PDF waits for the statement EEM generates in Stage 7. The four Thermal Plant sample PDFs wait for Stage 7 as well, for the same reason: the cost centers' bills are allocated there. How an image reaches a bill on an imported account (Bill Entry lists only Bill Entry accounts) is not confirmed; record what works.

Record: nothing generated.

Check: `images_attached`, `notes_present`.

Step 5. Approval

Where: the bill's status chip and status history in `#/BillEntry`; the Validate action on Bill Validation.

Enter: 05-approval.yaml.

Order: every bill already carries a Status_Code 1 (Bill Entry) history row from being saved or imported. First, with the job still off, open the Sierra Gas and Helios bills that post to fiscal periods 3, 6, 9 and 12 (8 of an account's 24, exactly a third; 104 in all with Library's 7 and the exception) and click Validate; also validate the Student Center estimate. Then, as the fixture in the second record, sign in as a bill-entry-only user and confirm she cannot do the same. Last, set SysConstants IsValidateBillsJob to 1 and let the built-in BILL_VALIDATION task run once: every bill without a finding reaches Status_Code 2 by itself (that is how the imported accounts are approved, under EEMValidate_Job), and a bill with a finding is put back to Status_Code 1 - the estimate included.

Record: nothing generated; the status change is the record.

Check: `approvals_by_role`, and now `validation_findings_expected`.

Stage 5 is done when

- Bill Processing and Bill Tracking's core reports render with the phase-0-1 relationships holding on every bill: charge lines sum to the bill total, Valley Electric's TOU lines sum to total kWh, River City's tiers partition total kgal, Helios's credit line is negative, Sierra Gas and Valley Electric's tax lines are a percent of the subtotal before them (the Central Plant's, from the September 2024 bill, leave the VED-BCT credit and fee out of that subtotal), and site/building rollups equal the sum of their bills.
- Bill Validation rows exist at Global, Company, Group and Account scope on both tabs, 42 of the 48 tests are turned on, and each of the 39 exercised tests has a bill that trips it and one that does not (`test_catalog`); the Ridgeline generator override switches off both change tests.

- Two Bill Importer sources are configured and the fiscal-period and bad-data samples are rejected with the product's own messages, not silently dropped.
- Every account that takes bills in this stage has 24 monthly bills for 2024-2025 (23 for Sierra Gas Library, 25 for Sierra Gas Student Center), the Ridgeline generator account has 17 (August 2024 to December 2025), 1,049 in all; the six named fixtures behave as described; nineteen bill images and five notes are attached (the other five images join the Thermal Plant and Ridgeline bills in Stage 7); every bill's Total Due is the expected one, 24 of them printed on sample PDFs and the rest from the generator's feed; and the Bill Entry accounts' third has been approved by hand before the job approved the rest.
- Each generator credit has left its bill and landed once, sign reversed, on the Central Plant's next one: 16 matched pairs, with the last generator bill's (December 2025) still in transit, and Ridgeline's own account has no bill until Stage 7.

Run `checks.sql` beside this chapter against the instance database to see each of these as a result set.

Findings

- **Approval is the Bill Validation function, not Bill Administration.** The API endpoint that advances a bill to QC Approval - `BillEditController.ValidateBill/ValidateBills` (`src/Web/EEMSuite.Web.Endpoints/Controllers/BillEditController.cs`) - is gated by `[EEMPermission(Function, "Bill Validation")]`, and the permission catalog marks "Bill Administration" as a legacy, unmapped entry. Stage 1 now grants Bill Validation at level 1 through the shared EEM Company Setup group (eem_company_setup) and gives bill-entry groups none, so the approver-versus- entry test in Step 5 runs against the function the product checks. Product finding 2 records the model change (closed by design).
- **The Bill Importer app has no "add provider" screen.** A provider is a `Schemas/<code>.xml` file (documented inline in `Schemas/CRC.xml`) plus a code in `appsettings.json`'s `BillImporter:Utilities` list (`BillImporterMainWindowViewModel._TryLoadProvider`), both installed alongside the app rather than entered on any screen. Treated here as a one-time customization step, not a gap - see product findings if a same-instance "add a provider" screen is wanted later.
- **Bill Service Type is three values, not four.** The Phase 3 AP/GL plan described a four-way flag (No Upload / AP / GL / AP and GL). The field it actually maps to, `Global_Type_Code` where `Table_Code='BA'`, has three rows in this installed instance: 0 No AP Upload, 1 AP Upload Only, 2 Bill Payment (verified by direct query). There is no GL-only or AP-and-GL value; GL coding (the Liability Account and Default Expense Account fields) is present on every billing account regardless of its upload flag. Stage 6 reconciles the AP/GL plan's rules against the real three values.
- **A bill item has no unit of its own (Ridgeline).** The flat BIF item the Bill Importer reads is Meter, Descriptor, Estimate, Quantity and Amount, and the bill item it creates stores a commodity, a quantity and an amount, so a quantity takes the unit of the commodity it maps to. Valley Electric's generator statement prints the export in MWh beside kWh; a figure mapped to a commodity of the other unit is stored a thousand times off, and nothing flags it. Northlake carries the export as dollars only, so nothing here depends on it.
- **A bill line cannot say it is generation (Ridgeline).** The generation credit has an export quantity and a negative amount, but no commodity marks energy as exported rather than used (product finding 44 for meters and points; bills have the same gap). Mapped to a kWh commodity it adds the export to the building's use; mapped to Misc Chgs/Credits it keeps the dollars and drops the kWh from the bill. Northlake takes the second, and the export stays in the received interval channel.
- **The validation job approves, and un-approves, on its own.** The BILL_VALIDATION task (`uaBillValidate_Job`) runs the Batch Test rules over every saved or imported bill: a bill with no finding gets

Status_Code 2 (written by EEMValidate_Job), a bill with a finding is put back to 1. The planned "one third approved" therefore only holds on the Bill Entry accounts until the job is switched on, and a manual Validate on a bill with findings is undone by the job's next pass.

- **An imported bill is never validated by Bill Save.** Only the job's Batch Test tab sees it, so Batch Test has to carry the same rules as Bill Save.
- **There is no Estimate checkbox and no supersedes link.** Bill Type (0 normal, 1 estimated, 2 accrual) is the only estimate marker an entered bill has, the importer drops the BIF's Estimate column, and `Bill.refBillID` is read and written by nothing in the product.
- **A Bill Allocation account cannot be opened in Bill Entry.** The Thermal Plant's bills can only arrive through allocation (Stage 7).
- **Bill Importer utility rows are per provider and company text** (`BillImportPro_Utility` is unique on description, provider and company name), so three companies for each of two providers is up to six rows, not two.

None of the above blocks entering this stage's bills - they are recorded so Stage 6 and any later permission cleanup start from what is actually installed.